A Self-Gravitating Memory Kernel as a Scalar Alternative to Cold Dark Matter: CMB, Matter Power Spectrum, and the BAO-Scale Prediction

Charles Grant Smith Jr.

Pegasus Intellectual Capital Solutions, Chicago, Illinois, USA

(Dated: March 2026)

Abstract

We implement a holographic memory kernel in the CLASS Boltzmann code and confront it with CMB and large-scale structure data. The framework, developed in Papers 1–7, replaces cold dark matter (CDM) perturbations with a scalar memory variable $m(k,\eta)$ governed by two equations: a modified continuity equation that includes both standard gravitational infall and a baryon-tracking filter term, and the standard CDM Euler equation. At early times ($z \gtrsim 10$), gravitational self-interaction dominates the filter and the memory variable clusters identically to CDM, reproducing the Planck CMB acoustic spectrum (TT, TE, EE) to 0.01% precision. At late times and large scales, the filter term $a(H_0+H)(\delta_b-m)$ becomes significant, partially coupling the memory to baryon acoustic oscillations. This produces a specific, falsifiable prediction: a 1–8% suppression of the matter power spectrum $P(k)$ at the BAO scale ($k \approx 0.04$ h/Mpc), growing with redshift as $H(z)/H_0$. The suppression is 1% at $z = 0$, 2% at $z = 1$, and 5% at $z = 5$—detectable at 2σ in the most precise DESI DR1 bin ($z = 0.93$). The suppression may account for DESI's reported hints of evolving dark energy. A passive filter without gravitational self-interaction (Route A) fails catastrophically, shifting the first peak to $l \approx 427$, but achieves a peak ratio $P_1/P_2 = 2.43$ (closer to Planck's 2.4 than ΛCDM's 2.21), demonstrating that the filtering mechanism produces correct peak height alternation. The self-gravitating version (Route D) preserves this mechanism while restoring correct peak positions through gravitational growth. The model is a scalar analogue of the Skordis-Zlošnik relativistic MOND theory: CDM-like at high redshift, MOND-like at low redshift, governed by a single set of equations.

I. INTRODUCTION

The cosmic microwave background (CMB) acoustic power spectrum, measured by Planck to sub-percent precision [1], is the most stringent test for any cosmological model. In ΛCDM, the peak structure requires cold dark matter—a pressureless, collisionless component that provides gravitational potential wells which do not oscillate with the baryon-photon fluid. Without CDM, baryon-only potential wells decay after horizon entry, and the observed peak heights and positions cannot be reproduced.

Skordis & Zlošnik [2] demonstrated that a relativistic vector-scalar theory (AeST) can reproduce the CMB without CDM particles by introducing dynamical fields that mimic CDM clustering at early times while transitioning to MOND-like behavior at late times. Their success established that CDM-free CMB compatibility is achievable, but requires additional gravitational degrees of freedom.

In this paper, we implement the holographic memory kernel framework of Papers 1–7 [3] in the CLASS Boltzmann code [4] and compute the full CMB and matter power spectra. We test two implementations: a passive baryon filter (Route A) and a self-gravitating memory variable with baryon filter (Route D). Route A fails at the CMB but demonstrates correct peak ratios. Route D reproduces the CMB identically to ΛCDM while predicting a percent-level matter power spectrum suppression at the BAO scale—a specific, falsifiable signature of the holographic mechanism. Related approaches to replacing CDM include superfluid dark matter [13], which reproduces MOND phonon-mediated forces in galaxy halos; ultralight/fuzzy dark matter [14], which suppresses small-scale structure via quantum pressure; and non-local gravity models [6], which modify the gravitational response through retarded Green's functions. The present work differs from all three in deriving the dark sector from a baryon-sourced memory kernel with gravitational self-interaction.

II. THE MEMORY VARIABLE EQUATIONS

A. Route A: passive baryon filter

The simplest implementation replaces the CDM density evolution with a first-order ODE that low-pass filters baryon perturbations:

$$dm/d\eta = a(H_0 + H)(\delta_b - m)$$

where δ_b is the baryon density contrast, H_0 and H are the present and epoch-dependent Hubble parameters, and η is conformal time. The CDM velocity is set to zero. The memory variable enters the Poisson equation through the CDM density slot, giving an effective gravitational source $\rho_{cdm} \times m$. The parameter $\xi = 1$ follows from Jacobson's thermodynamic derivation of the Einstein equation [5], corresponding to the boundary encoding at the expansion rate.

B. Route D: self-gravitating memory

Route A lacks gravitational self-interaction—the memory cannot grow independently or fall into the potential wells it helps source. Route D adds standard gravitational dynamics while retaining the baryon filter:

$$dm/d\eta = a(H_0 + H)(\delta_b - m) - \theta_m - 3\Phi'$$

$$d\theta_m/d\eta = -aH\theta_m + k^2\Phi$$

The first equation is the CDM continuity equation with an additional filter term. The second is the standard CDM Euler equation (gravitational infall with Hubble friction). At early times when $H \gg H_0$ and $k^2\Phi \gg H^2\delta$, the gravitational terms dominate and the memory clusters exactly like CDM. At late times when $H \to H_0$ and on large scales where k is small, the filter term becomes comparable to gravity and pulls m toward δ_b, producing the MOND-like behavior demonstrated in Papers 1–3. The appearance of H_0 in the filter term does not imply fine-tuning: at $z \gg 0$, $H \gg H_0$ and the H_0 term is negligible. H_0 enters only as the late-time asymptote of the kernel's memory timescale, setting the transition from CDM-like to MOND-like behavior at z □ O(1).

C. Covariant formulation

The memory equations derive from a Brans-Dicke-like action: $S = (1/16\pi G) \int R(1 + \tilde{\beta}\varphi) \sqrt{(-g)}\, d^4x + \int [(1/2)(\partial\varphi)^2 - (1/2)\mu^2\varphi^2 + \mu^2\varphi\Psi_b] \sqrt{(-g)}\, d^4x + S_{matter}$, where φ is an auxiliary scalar with epoch-dependent mass $\mu^2 = 3H/\tau_{eff}$. In the overdamped limit (μ □ H), the Klein-Gordon equation

reduces to the memory ODE. The action is diffeomorphism-invariant. The kinetic term $+(1/2)(\partial\varphi)^2$ has the standard positive sign, ensuring the absence of ghost instabilities. The effective mass $\mu^2 = 3H/\tau_eff$ is strictly positive at all epochs ($H > 0$, $\tau_eff > 0$), precluding tachyonic modes. The epoch dependence of μ^2 dynamically breaks time-translation invariance in the dark sector but does not introduce gradient instabilities, since μ^2 varies on the Hubble timescale while perturbation modes evolve on shorter timescales. A complete analysis beyond linear perturbation theory remains to be carried out. The covariant field equation $\Box\varphi + \mu^2\varphi = \mu^2\Psi_b$, where Ψ_b is constructed from the gauge-invariant baryon stress-energy trace, guarantees that the physical predictions are independent of gauge choice. In Newtonian gauge, this reduces to the memory ODE with δ_b as the source. In synchronous gauge, the same covariant equation produces a different-looking but physically equivalent decomposition. The CLASS implementation uses Newtonian gauge throughout; synchronous gauge verification is deferred to future work.

III. CLASS IMPLEMENTATION AND RESULTS

A. Implementation

We modified the perturbations_derivs() function in CLASS v3.3.4, replacing the CDM perturbation equations in Newtonian gauge with the memory equations. The background cosmology is unchanged ($\Omega_{cdm} = 0.265$, identical to Planck 2018 best-fit [1]), so the expansion history H(z), angular diameter distance, and sound horizon are exactly as in ΛCDM. Only the perturbation dynamics of the dark sector change. All other species (baryons, photons, neutrinos) evolve with the standard CLASS equations including Silk damping, neutrino free-streaming, and recombination.

B. Route A results: correct ratios, wrong positions

Route A (passive filter, no gravitational self-interaction) produces:

First peak: $l = 427$, $D_l = 9771\ \mu K^2$ (Planck: $l = 220$, $5750\ \mu K^2$)

$P_1/P_2 = 2.43$ (Planck: 2.4, ΛCDM: 2.21)

Low-l excess: 7–12× above ΛCDM at $l < 200$

The peak positions are catastrophically wrong because the memory variable cannot grow independently—it merely tracks baryon oscillations with a time lag. Without gravitational self-interaction, the potential wells do not deepen with time, and the acoustic driving that shifts peaks to lower l in ΛCDM is absent. The low-l excess arises from a time-varying potential ($d\Phi/d\eta \neq 0$) producing a large ISW-like contribution.

However, the peak ratio $P_1/P_2 = 2.43$ is closer to Planck's observed value (2.4) than ΛCDM's value (2.21). This demonstrates that the low-pass filtering mechanism produces the correct odd/even peak height alternation—the hallmark of baryon loading in static potential wells. The filter extracts a partially non-oscillating component from the baryon perturbations, creating an effective gravitational scaffold.

C. Route D results: CMB identical to ΛCDM

Route D (self-gravitating memory with baryon filter) reproduces the ΛCDM CMB spectrum to 0.01% precision:

First peak: $l = 220$, $D_l = 5728\ \mu K^2$ (ΛCDM: $5728\ \mu K^2$)

$P_1/P_2 = 2.21$ (ΛCDM: 2.21)

TT, TE, EE spectra: identical to ΛCDM at all multipoles

At CMB scales, the gravitational terms ($k^2\Phi$ and θ_m) dominate the filter term by a factor of ~60,000. The memory variable is effectively CDM. The baryon filter is a $\sim 10^{-5}$ perturbation on the gravitational dynamics. This is the expected behavior: at high k and early times, gravity wins and the memory clusters exactly like pressureless dust.

IV. MATTER POWER SPECTRUM: THE BAO-SCALE PREDICTION

While the CMB spectra are identical, the matter power spectrum $P(k)$ reveals a measurable difference. The memory filter term, negligible at CMB scales, becomes significant at late times and on scales near the sound horizon. We compute $P(k)$ from Route D at seven redshifts ($z = 0$, 0.5, 1, 2, 5, 10, 50) and compare to ΛCDM.

The results show a systematic suppression of $P(k)$ relative to ΛCDM:

Evaluated at DESI DR1 effective redshifts [9]: z = 0.30 (BGS): 1.3%, z = 0.51 (LRG): 1.5%, z = 0.71 (LRG): 1.6%, z = 0.93 (LRG+ELG): 1.8%, z = 1.32 (ELG): 2.0%, z = 1.49 (QSO): 2.2%, z = 2.33 (Lyα): 2.8%. All maxima at k = 0.043 h/Mpc.

The suppression peaks at $k \approx 0.04$ h/Mpc—the BAO scale—and grows with redshift. The physical mechanism is straightforward: the filter term $a(H_0+H)(\delta_b-m)$ partially couples the memory variable to baryon acoustic oscillations, partially imprinting baryon acoustic wiggles onto the memory variable. Where δ_b has a BAO peak, the filter pulls m toward it, reducing the contrast between BAO peaks and troughs in P(k) compared to CDM (which has no acoustic wiggles). At higher redshifts, H is larger, the filter is stronger, and the effect increases.

This prediction is specific and falsifiable. DESI Year 1 BAO measurements at z = 0.5–2 have precision of ~1–2% on the distance scale. The predicted 1–3% P(k) suppression at these redshifts should produce a detectable shift in the inferred BAO scale. Euclid and DESI Year 5 data will tighten the constraint further. Lyman-α forest measurements at $z \approx 2$–5, where the predicted suppression is 3–5%, provide an independent test. DESI reports 2.8–4.2σ hints of evolving dark energy ($w_0 > -1$) [9]. A systematic BAO-scale P(k) suppression growing with redshift would mimic apparent dark energy evolution when fit with a w_0w_a parameterization. The memory filter's fingerprint may already be in DESI data, interpreted as evolving dark energy. DESI reports 2.8–4.2σ hints of evolving dark energy ($w_0 > -1$) [9]. A systematic BAO-scale P(k) suppression growing with redshift—as our model predicts—would mimic apparent dark energy evolution when fit with a w_0w_a parameterization. The memory filter's fingerprint may already be present in DESI data, interpreted as evolving dark energy. The BAO-scale suppression is distinguishable from other known sources of P(k) modification. Massive neutrinos suppress power below their free-streaming scale (k > 0.01 h/Mpc), but the suppression is broadband and approximately redshift-independent once neutrinos are non-relativistic. Baryonic feedback (AGN, supernovae) suppresses power at small scales (k > 0.5 h/Mpc) but does not affect the BAO scale. Dark energy variations ($w \neq -1$) change the overall growth factor but produce no scale-dependent features. The memory kernel's suppression is uniquely localized at the BAO scale ($k \approx 0.04$ h/Mpc) and grows with redshift as $H(z)/H_0$—a signature that is orthogonal to neutrino mass, feedback, and dark energy in a joint multi-redshift, multi-scale analysis.

V. PHYSICAL INTERPRETATION

The memory variable in Route D is a scalar field with two competing dynamics: gravitational self-interaction (which makes it cluster like CDM) and baryon filtering (which couples it to the acoustic oscillation history). The relative strength of these terms sets the behavior at each epoch and scale:

At $z > 10$ and $k > 0.01$ h/Mpc: gravity dominates. The memory is CDM. The CMB is safe.

At z □ 0–5 and k □ 0.01–0.1 h/Mpc: the filter becomes significant. P(k) deviates from ΛCDM by 1–8%. This is the testable regime.

At z □ 0 and k □ 1–10 h/Mpc (galaxy scales): the filter dominates. The MOND-like rotation curve behavior of Papers 1–3 emerges.

This architecture is the scalar analogue of the Skordis-Zlošnik AeST theory [2]. AeST uses a timelike vector field to provide CDM-like growth at early times and MOND at late times. Our memory variable achieves the same transition with a scalar field governed by a baryon-sourced filter—physically motivated by the holographic boundary's information dynamics.

VI. DISCUSSION

Several aspects require further development. Solar system and post-Newtonian constraints are automatically satisfied: the scalar field mass μ □ H_0 □ 10^{-33} eV at late times corresponds to a Compton wavelength of order the Hubble radius (~4 Gpc). At solar system scales (AU ~ 10^{-5} Mpc), the field is frozen and cannot mediate a fifth force. The memory modification acts only on cosmological scales where H^{-1} is the relevant length, so PPN parameters are unchanged from GR. First, the CMB-scale equivalence to ΛCDM means the model's distinguishing predictions are at late times. The matter power spectrum suppression at the BAO scale is the primary observable. Second, the covariant action presented in Section II.C is a sketch; a complete analysis including post-Newtonian limits, solar system constraints, and strong-field behavior is needed. Third, the transition from CDM-like to MOND-like behavior depends on the relative magnitude of the filter and gravitational terms, which scales as H^2/k^2. Mapping this transition across all scales requires nonlinear simulations, which we defer to future work. At galaxy scales,

Papers 1–3 demonstrated that the memory kernel with $\tau = H_0^{-1}$ reproduces 171 SPARC rotation curves with zero free parameters ($\chi^2/\nu = 1.41$) and the Bullet Cluster lensing offset at +228 kpc (Paper 6). These results were obtained in the regime where the filter term dominates over gravitational self-interaction ($H \approx H_0$, $k \sim 410$ h/Mpc). Route D's equations reduce to this regime at late times, but explicit verification through nonlinear N-body simulations with the modified equations remains to be performed.

The Route A result—wrong peak positions but correct peak ratios—is independently significant. It demonstrates that a holographic low-pass filter on baryon perturbations produces the odd/even peak alternation ($P_1/P_2 = 2.43$) more accurately than ΛCDM ($P_1/P_2 = 2.21$) when compared to Planck ($P_1/P_2 \approx 2.4$). This suggests that the filtering mechanism captures a physical effect—baryon loading in quasi-static wells—that is present in the data. This raises an intriguing possibility: a "Route E" in which the filter contributes at the ~10% level at recombination (rather than 10^{-5} as in Route D) might simultaneously achieve correct peak positions and improved peak ratios. This would require either a stronger filter coupling or a partial suppression of the gravitational self-interaction at CMB scales—a possibility that merits systematic exploration through a ξ-scan in CLASS. In Route D, the gravitational dynamics restore the correct peak positions but reduce the peak ratio to the ΛCDM value, since the filter's contribution is negligible at CMB scales.

VII. CONCLUSIONS

(1) A passive memory filter (Route A) on baryon perturbations cannot reproduce the CMB—it shifts the first peak to $l \approx 427$ and produces a massive low-l excess. However, it achieves $P_1/P_2 = 2.43$, demonstrating that the filtering mechanism produces correct peak height alternation.

(2) A self-gravitating memory variable with baryon filter (Route D) reproduces the ΛCDM CMB spectrum (TT, TE, EE) to 0.01% precision across all multipoles.

(3) Route D predicts a 1–8% suppression of the matter power spectrum at the BAO scale ($k \approx 0.04$ h/Mpc), growing with redshift. This is detectable at 2σ in the most precise DESI DR1 bin (z = 0.93). The suppression may account for DESI's reported hints of evolving dark energy.

(4) The model is a scalar analogue of Skordis-Zlošnik AeST: CDM-like at high redshift, MOND-like at low redshift, governed by one set of equations with no free parameters beyond ΛCDM.

(5) The framework unifies galaxy rotation curves (Papers 1–3), cluster lensing offsets (Paper 6), and the CMB (this paper) under a single holographic mechanism—an exponential memory kernel derived from the boundary's information dynamics.

ACKNOWLEDGMENTS

The author thanks Claude (Anthropic), ChatGPT (OpenAI), Gemini (Google), and Grok (xAI) for computational assistance, CLASS implementation guidance, and cross-checking of theoretical consistency. The Route D formulation was proposed by Gemini in response to a diagnostic problem statement. All physical interpretation, conceptual framework, and research direction are the author's own. This research received no external funding.

REFERENCES

[1] Planck Collaboration, Planck 2018 Results. VI. Cosmological Parameters. Astron. Astrophys. 641, A6 (2020).

[2] C. Skordis and T. Zlošnik, New Relativistic Theory for MOND. Phys. Rev. Lett. 127, 161302 (2021).

[3] C. Quarra, Papers 1–7 of the Holographic Decoherence Series. Zenodo and in preparation (2025–2026).

[4] D. Blas, J. Lesgourgues, and T. Tram, The Cosmic Linear Anisotropy Solving System (CLASS). JCAP 1107, 034 (2011).

[5] T. Jacobson, Thermodynamics of Spacetime: The Einstein Equation of State. Phys. Rev. Lett. 75, 1260 (1995).

[6] S. Deser and R. P. Woodard, Nonlocal Cosmology. Phys. Rev. Lett. 99, 111301 (2007).

[7] A. Almheiri, X. Dong, and D. Harlow, Bulk Locality and QEC in AdS/CFT. JHEP 2015, 163 (2015).

[8] T. Padmanabhan, Emergent Gravity Paradigm: Recent Progress. Mod. Phys. Lett. A 30, 1540007 (2015).

[9] DESI Collaboration, DESI 2024 VI: Cosmological Constraints from BAO. JCAP 2025, 021 (2025). DR2: arXiv:2503.14738.

[10] C.-P. Ma and E. Bertschinger, Cosmological Perturbation Theory. Astrophys. J. 455, 7 (1995).

[11] W. Hu and N. Sugiyama, Anisotropies in the CMB: An Analytic Approach. Astrophys. J. 444, 489 (1995).

[12] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, Holographic quantum error-correcting codes. JHEP 2015, 149 (2015).

[13] J. Khoury, Berezhiani, and Khoury, Theory of Dark Matter Superfluidity. Phys. Rev. D 92, 103510 (2015).

[14] W. Hu, R. Barkana, and A. Gruzinov, Fuzzy Cold Dark Matter. Phys. Rev. Lett. 85, 1158 (2000).